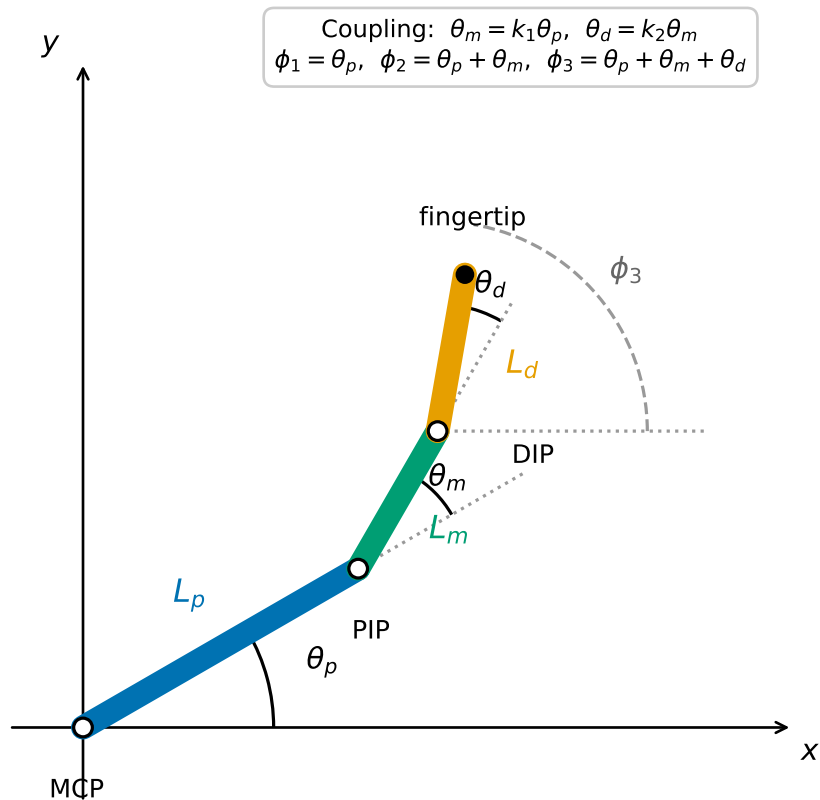
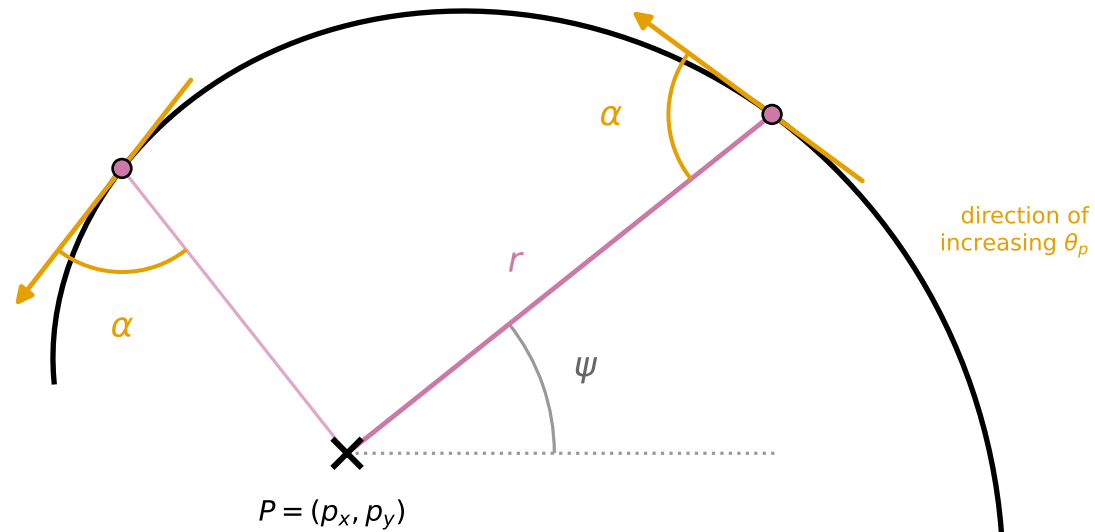


(A) Forward kinematics and the coupling constraint



(B) Spiral geometry about the pole  $P$



Equiangular spiral:  $r = a e^{\mu \psi}$ ,  $\mu = \cot \alpha$   
 $\alpha$  constant along the curve  $\Leftrightarrow$  exact spiral